ECE330 Quiz4

完整题目、你的完整回答与对应解答

共3题，逐题保留实际网页记录与上传手写原图。

每题按原页连续截图、完整手写图、解答的顺序排列。

长网页分段连续展示，相邻页留有重叠，防止截断文字。

手写图使用完整上传图片，未裁去纸面上的作答。

历史上传图片不同的版本均保留；相同图片不重复放大。

截图与数值记录来自你的已完成测验页面。

解答为独立解析，系统答案与反馈在原页截图中保留。

截图日期 2026年11月4日

Quiz4 第1题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 1

NOTE: Full credit (100%) for this question is 7 points: all for the correct numerical answer(s).

(7 pts) The flux linkage in an energy-conservative electromechanical system is given by

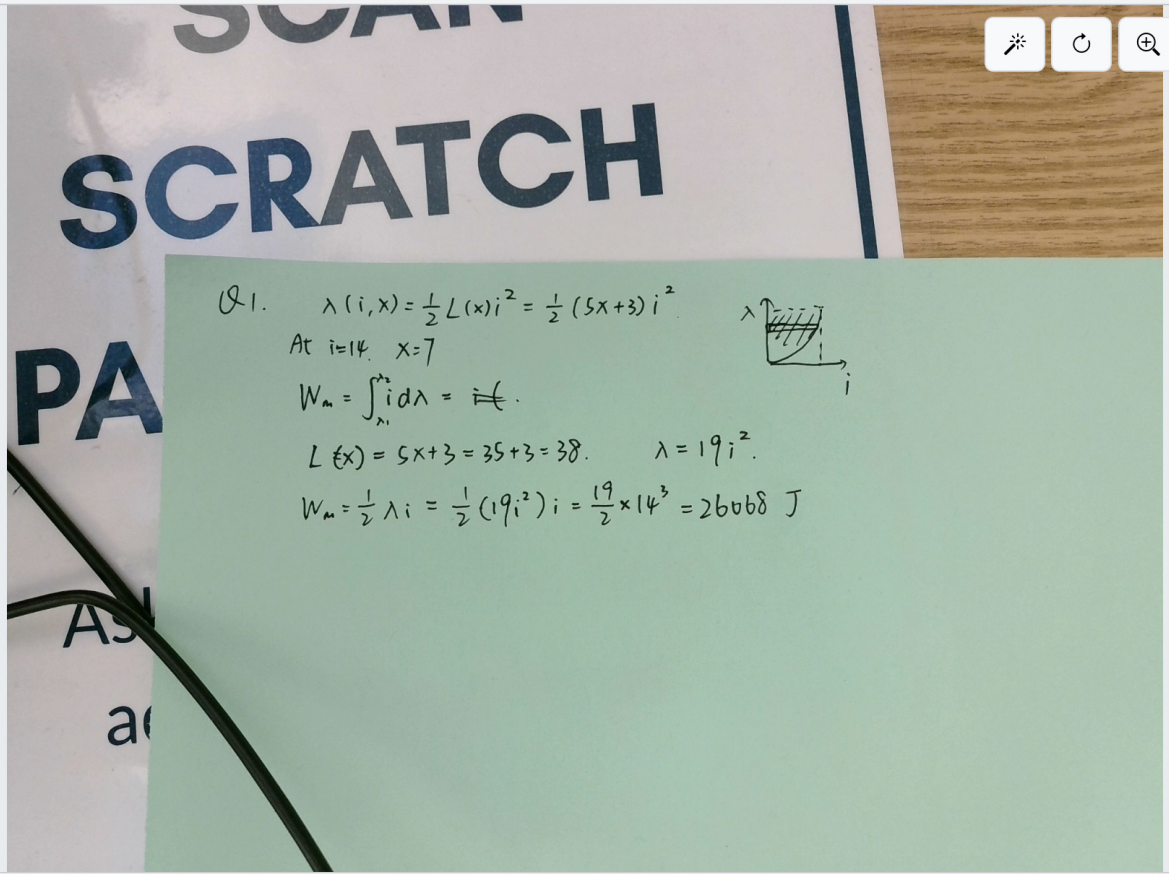
$$\lambda(i,x) = \frac{1}{2}L(x)i^2$$

where $L(x) = 5x + 3$ H/A. Here, x is the size of the gap in m, and i is the current applied to the coil in A.

Compute the energy (in Joules) in the system at $i = 14$ A and $x = 7$ m.

Wm [J]= 26068 ✕ 0%

Image capture



Correct answer

Wm [J]= 34757.3333333

Submitted answer

yitaoma2@illinois.edu submitted at 2026-11-03 19:06:38 (CST) 0%

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Quiz4 第1题 原始页面 2/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

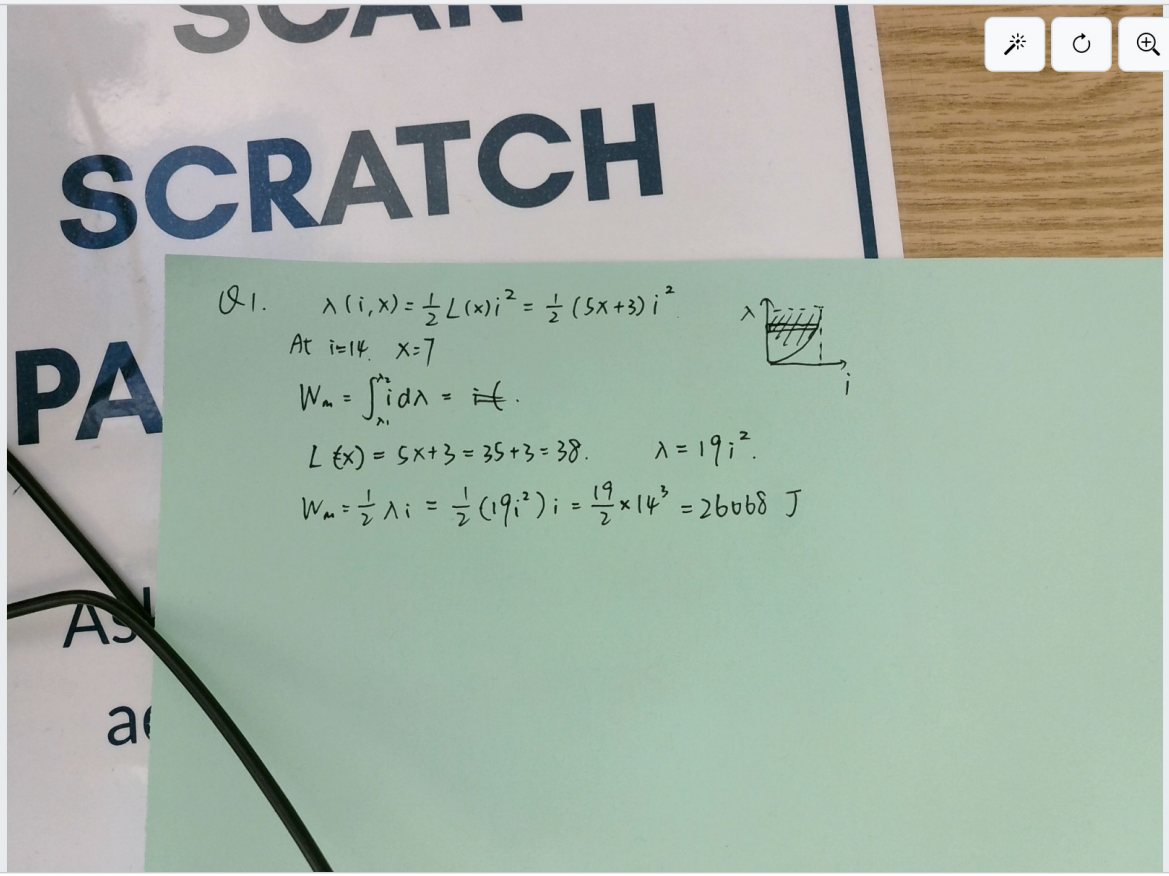
Submitted answer

yitaoma2@illinois.edu submitted at 2026-11-03 19:06:38 (CST) 0%

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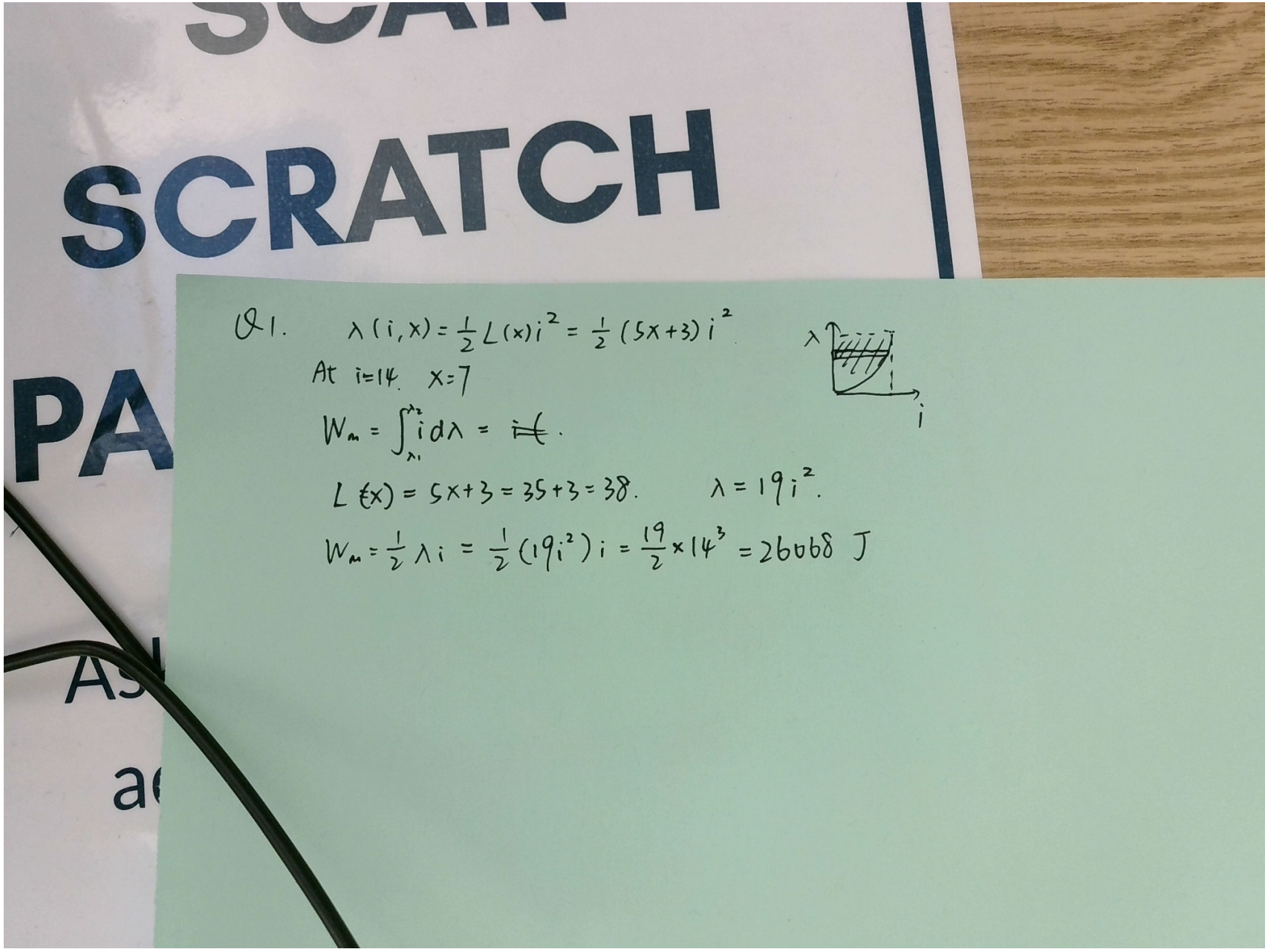
Wm [J]= 26068 | Your answer does not have precision within the specified relative tolerance. ✕ 0%

Image capture



Quiz4 第1题 你的完整手写回答 1/1

当前最终回答原图



Quiz4 Q1 | Full uploaded image | 2000 x 1500 px

第1题 非线性磁链的磁场能量

中文题意

保守机电系统满足 $\lambda(i,x)=\frac{1}{2}L(x)i^2$，其中 $L(x)=5x+3$，单位为 H/A；x 为以米计的气隙，i 为以安培计的线圈电流。求 $i=14\,\mathrm{A}$、$x=7\,\mathrm{m}$ 时的磁场能量 W_m，单位 J。

推导与结果

这里要求的是磁场能量，即在位置固定时对磁链积分 $W_m=\int i\,d\lambda$。题给磁链正比于电流的平方，不能套用线性模型的 $\frac{1}{2}i\lambda$。先固定 $x=7$，求出本构系数：

$$L(7)=5(7)+3=38\ \mathrm{H/A}.$$

将题给磁链对电流求导，得到积分中的磁链增量：

$$d\lambda=L(x)i\,di.$$

把这个表达式放回能量定义，积分变量用 $\hat{i}$ 区分于最终电流：

$$W_m=\int_0^{14}\hat{i}\,[38\hat{i}\,d\hat{i}]=38\left[\frac{\hat{i}^3}{3}\right]_0^{14}$$

$$=\frac{38(14)^3}{3}=34757.3333\ \mathrm{J}.$$

核对：$\lambda=\frac{1}{2}(38)(14)^2=3724\,\mathrm{Wb\cdot turn}$；共能 $W'_m=38(14)^3/6=17378.6667\,\mathrm{J}$。两者之和为 $i\lambda=52136\,\mathrm{J}$。你填入的 $26068\,\mathrm{J}$ 对应 $\frac{1}{2}i\lambda$，这一步把非线性本构当成了线性本构。系统显示的正确值为 $34757.3333333\,\mathrm{J}$。

最终英文作答版

At fixed $x=7\,\mathrm{m}$, $L=5(7)+3=38\,\mathrm{H/A}$ and $d\lambda=38i\,di$. Therefore,

$$W_m=\int_0^{14} i\,d\lambda=\int_0^{14}38i^2\,di=\frac{38(14)^3}{3}=34757.3333\ \mathrm{J}.$$

The expression $W_m=\frac{1}{2}i\lambda$ does not apply because λ is quadratic in i.

Quiz4 第2题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 2

NOTE: Full credit (100%) for this question is 6 points: 3 points for the correct numerical answer(s) and 3 points for the correct conceptual answer. Points for the conceptual answer will be added manually.

The flux linkage for a given electromechanical system is given as $\lambda = \frac{Ai}{x - x_0}$ where $A = 0.059$ Hm and $x_0 = 0.01$ m

	a	b	c	d	a
i	0	i_b	i_c	0	0
λ	0	5	λ_c	0	0
x	0.038	0.038	0.022	0.022	0.038

The system is operated in a closed cycle $a \to b \to c \to d \to a$. x is held constant along $a \to b$ and $c \to d$ and i is held constant over $b \to c$ and $d \to a$.

(3 pts) Plot the system on a $\lambda - i$ plane. **Submit your answer on scratch paper.**

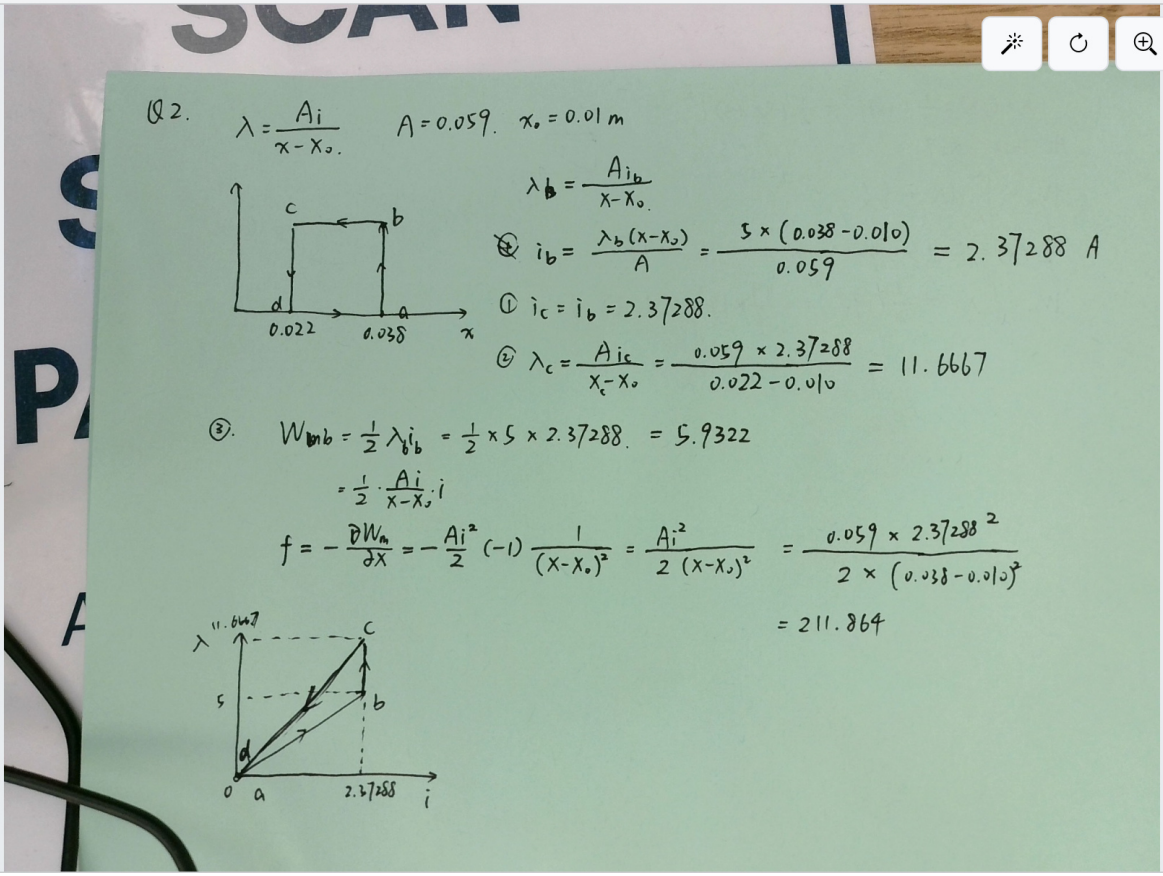
(3 pts) What is i_c and λ_c?

i_c[A]=	2.37288	?	✓ 100%
λ_c[Wb turns]=	11.6667	?	✓ 100%

What is energy stored in the coupling field and the force of electric origin at point b?

W_{m_b}[J]=	5.9322	?	✓ 100%
f_b^e[N]=	211.864	?	✗ 0%

Image capture



Quiz4 第2题 原始页面 2/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠



Correct answer

i_c [A]= 2.37288135593 λ_c [Wb turns]= 11.6666666667 W_{m_b} [J]= 5.93220338983 f_b^e [N]= −211.86440678

Feedback from the Course Staff Hide

Conceptual: Mostly incorrect answer (-2 pts)

Submitted answer
yitaoma2@illinois.edu submitted at 2026-11-03 19:17:43 (CST)
manual grading: 33%
auto-grading: 75%
Hide

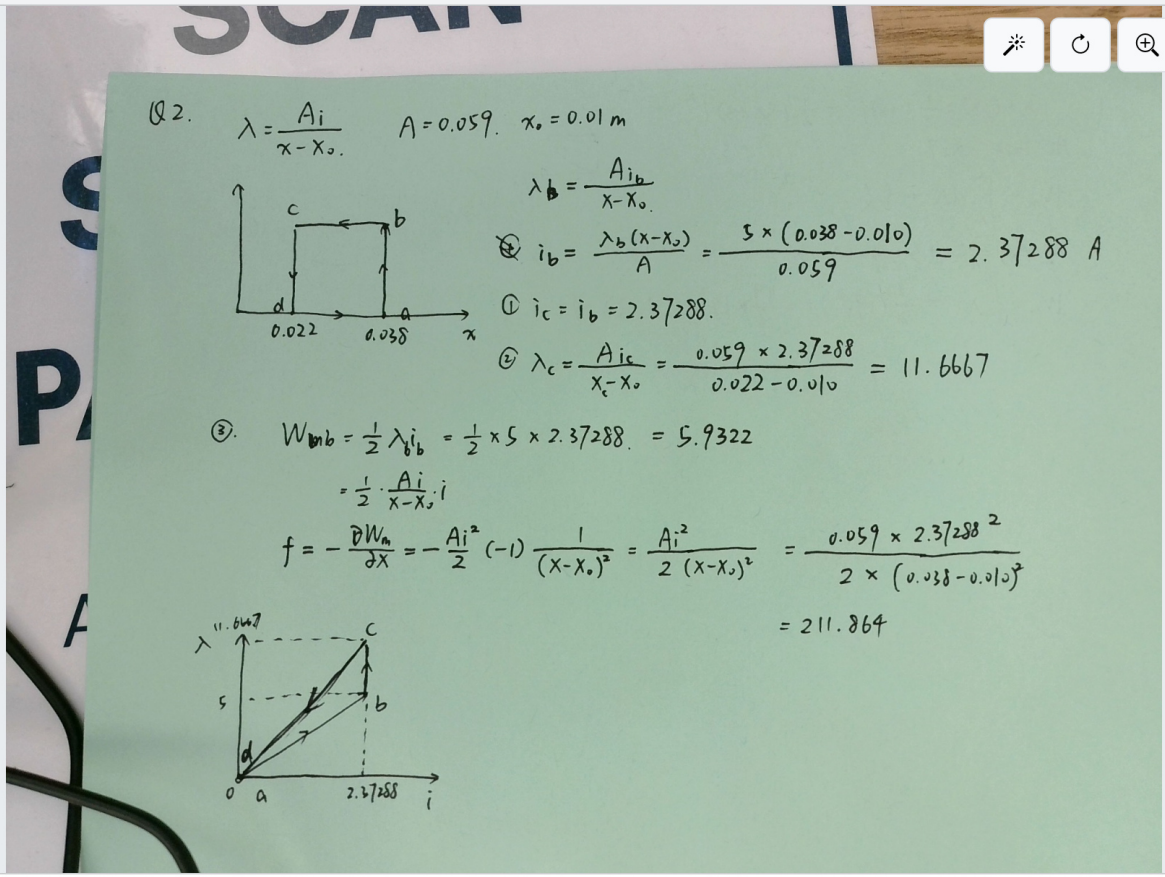
i_c [A]= 2.37288 | The correct answer used for grading was 2.3728813559322033. ✓ 100%

λ_c [Wb turns]= 11.6667 | The correct answer used for grading was 11.666666666666666. ✓ 100%

W_{m_b} [J]= 5.9322 | The correct answer used for grading was 5.932203389830508. ✓ 100%

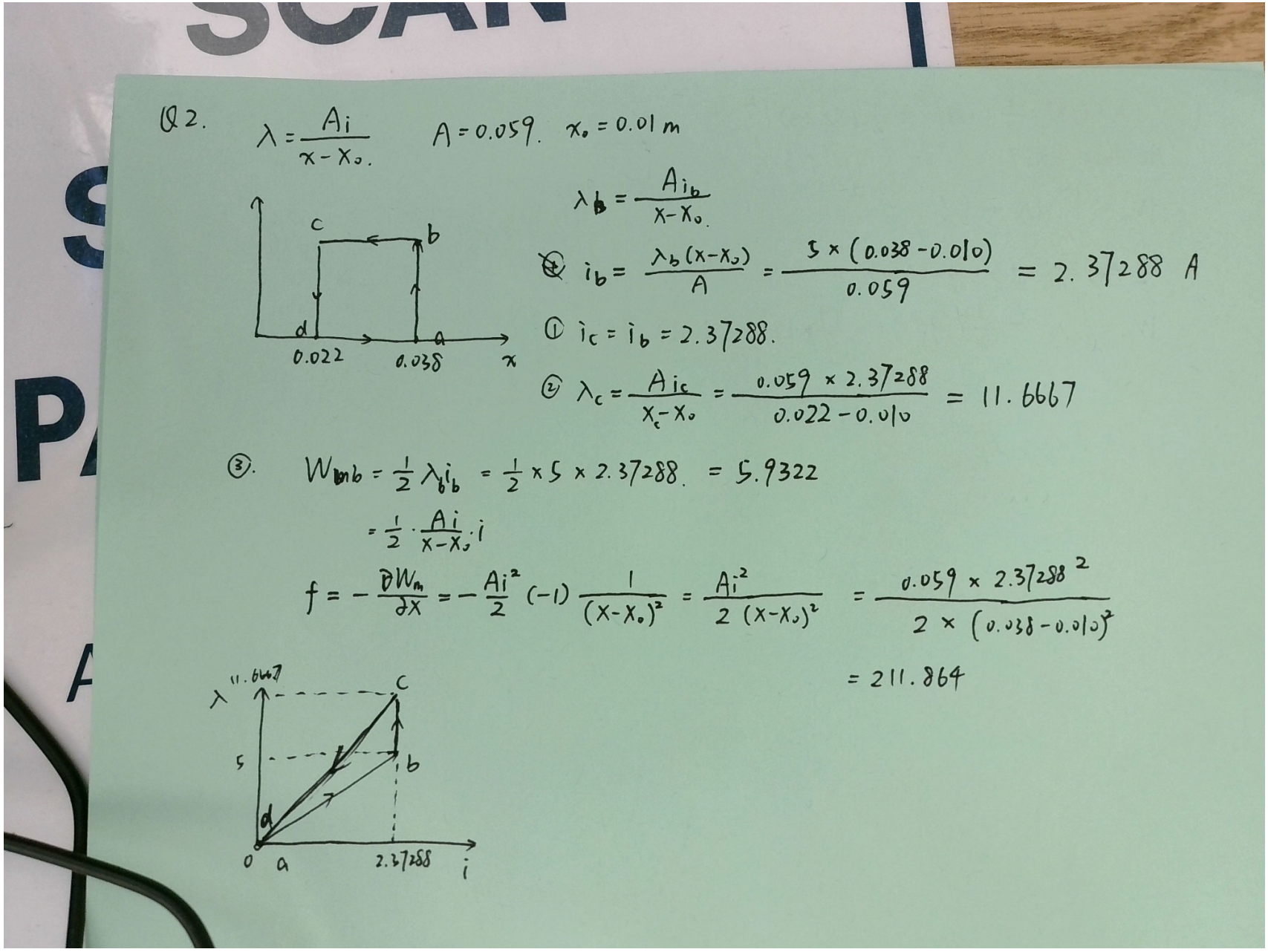
f_b^e [N]= 211.864 ✗ 0%

Image capture



Quiz4 第2题 你的完整手写回答 1/1

当前最终回答原图



Quiz4 Q2 | Full uploaded image | 2000 x 1500 px

第2题 闭合循环与固定磁链求力

中文题意

系统满足 $\lambda = Ai/(x - x_0)$，其中 $A = 0.059\,\mathrm{H\,m}$、$x_0 = 0.010\,\mathrm{m}$。按 $a \to b \to c \to d \to a$ 运行：$a \to b$ 和 $c \to d$ 固定位置，$b \to c$ 和 $d \to a$ 固定电流。状态表给出 $a: (i, \lambda, x) = (0, 0, 0.038)$，$b: (i_b, 5, 0.038)$，$c: (i_c, \lambda_c, 0.022)$，$d: (0, 0, 0.022)$，最后回到 a。在同一 λ–i 平面画循环，求 i_c、λ_c，以及 b 点的磁场能量和电磁力。

先由恒流段连接状态

先用 b 点的磁链和位置解电流，再用 $b \to c$ 恒流把这个电流带到 c 点：

$$i_b = \frac{\lambda_b(x_b - x_0)}{A} = \frac{5(0.038 - 0.010)}{0.059} = 2.37288135593\ \mathrm{A}.$$

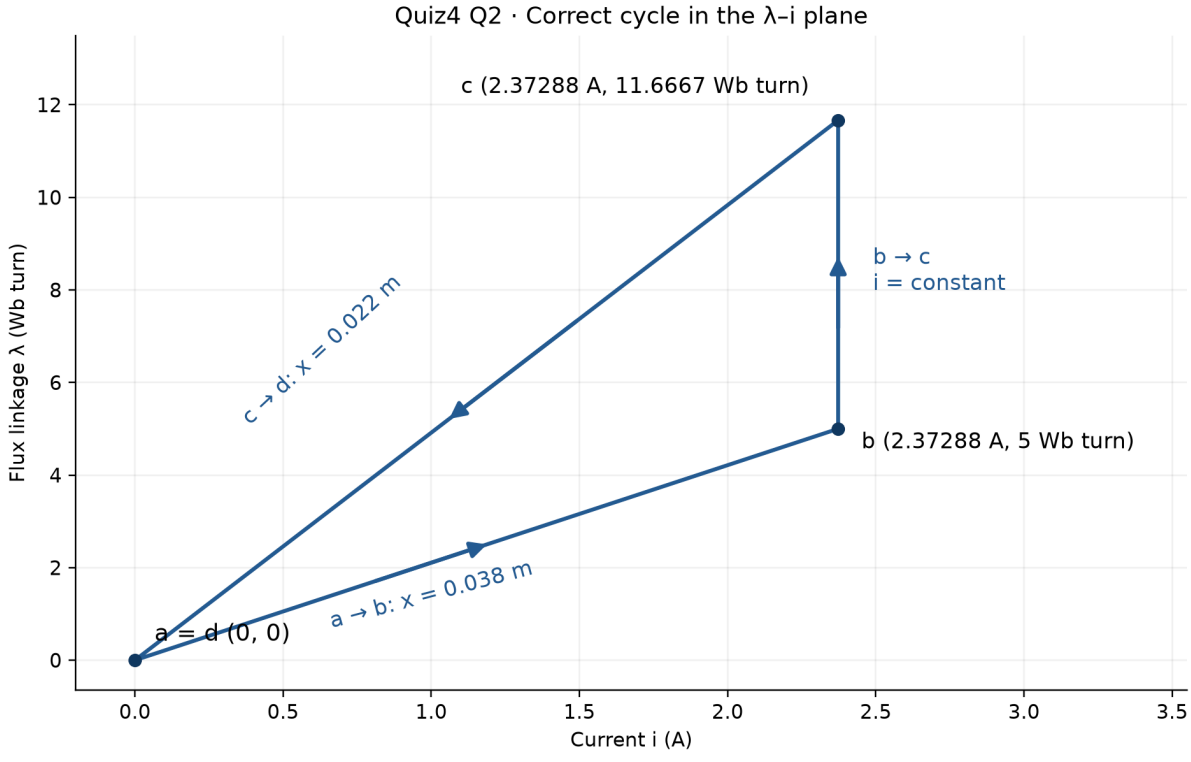
$$i_c = i_b = 2.37288135593\ \mathrm{A}.$$

将这个电流放回 c 点本构：

$$\lambda_c = \frac{0.059(2.37288135593)}{0.022 - 0.010} = 11.6666666667\ \mathrm{Wb \cdot turn}.$$

循环图与方向

横轴为电流 i，纵轴为磁链 λ。固定位置时是经过原点的直线；$b \to c$ 恒流，所以是竖直线。a 与 d 虽然位置不同，在这个平面都落在原点；$d \to a$ 不会产生另一条可见的线段。本问须手画或附上此图。



磁场能量与带符号电磁力

固定位置后，这道题的磁链对电流是线性的，所以能量可写为 $\frac{1}{2}i\lambda$：

$$W_{m,b} = \frac{1}{2} i_b \lambda_b = \frac{1}{2}(2.37288135593)(5) = 5.93220338983\ \text{J}.$$

求力时把能量写成磁链和位置的函数。由 $i = \lambda(x - x_0)/A$，对磁链积分：

$$W_m(\lambda, x) = \int_0^{\lambda} \frac{\widehat{\lambda}(x - x_0)}{A}\, d\widehat{\lambda} = \frac{\lambda^2 (x - x_0)}{2A}.$$

电磁力是固定磁链时能量对位置的负偏导。这里先固定 λ 求导，再代入 b 点：

$$f_b^e = -\left.\frac{\partial W_m}{\partial x}\right|_{\lambda} = -\frac{\lambda_b^2}{2A} = -\frac{5^2}{2(0.059)} = -211.864406780\ \text{N}.$$

负号表示力沿 x 减小的方向。你填写的电流、磁链和能量均与系统答案一致；力的大小一致，但输入少了负号。系统概念题反馈为“Mostly incorrect answer (−2 pts)”；此处保留该反馈原文，不推定更细的评分原因。

最终英文作答版

Since $b \to c$ has constant current,

$$i_c = i_b = \frac{5(0.038 - 0.010)}{0.059} = 2.37288135593\ \text{A},$$

$$\lambda_c = \frac{0.059(2.37288135593)}{0.022 - 0.010} = 11.6666666667\ \text{Wb} \cdot \text{turn}.$$

In the λ–i plane, draw $a = (0, 0)$, $b = (2.37288136, 5)$, $c = (2.37288136, 11.6666667)$, and $d = (0, 0)$, with arrows $a \to b \to c \to d$. The path $d \to a$ remains at the origin. Copy or attach the labeled cycle diagram above.

$$W_{m,b} = \frac{1}{2}(5)(2.37288135593) = 5.93220338983\ \text{J}.$$

$$W_m(\lambda, x) = \frac{\lambda^2 (x - x_0)}{2A}, \qquad f_b^e = -\left.\frac{\partial W_m}{\partial x}\right|_{\lambda} = -\frac{5^2}{2(0.059)} = -211.864406780\ \text{N}.$$

The force acts in the negative x direction.

Quiz4 第3题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 3

NOTE: Full credit (100%) for this question is 7 points: all for the correct numerical answer(s).

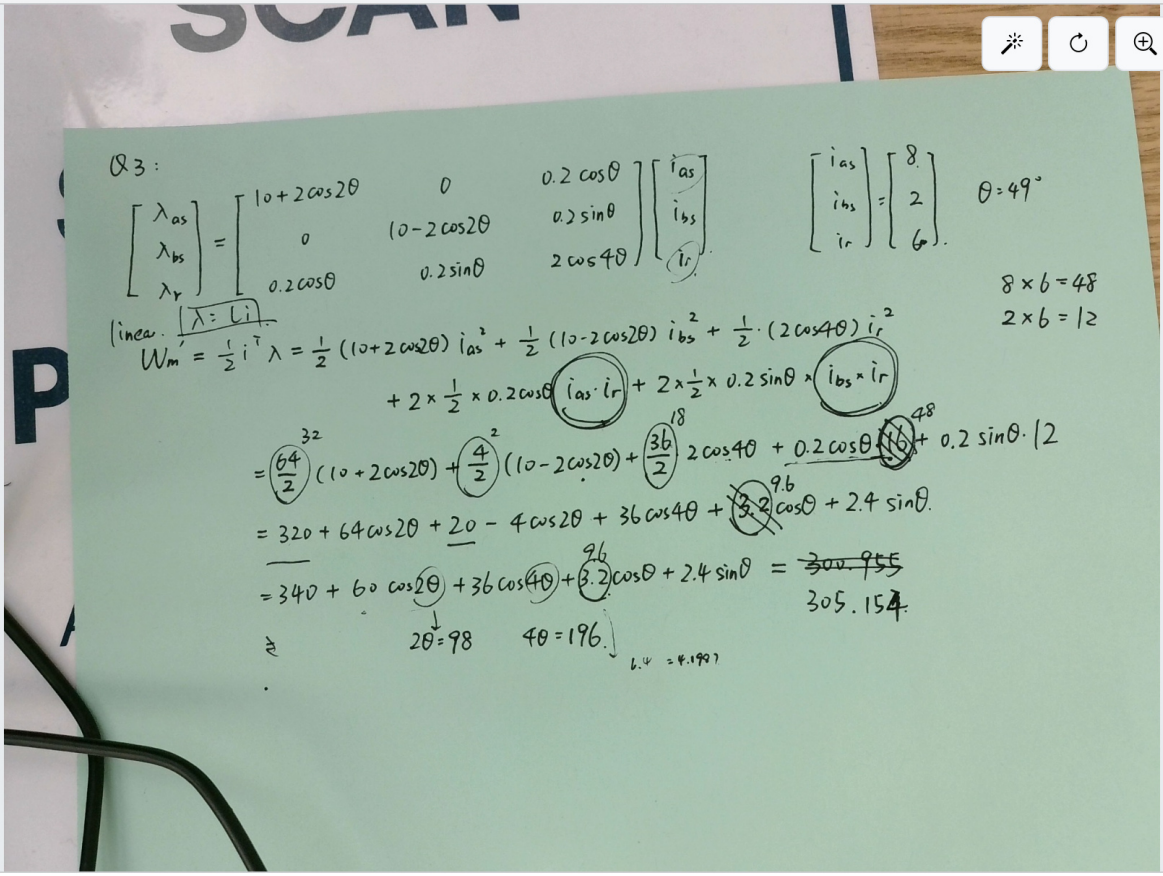
(7 pts) An electromechanical system has three electrical and one mechanical ports. The $\lambda - i$ relations are:

$$\begin{bmatrix}\lambda_{as}\\ \lambda_{bs}\\ \lambda_r\end{bmatrix} = \begin{bmatrix}10+2\cos(2\theta) & 0 & 0.2\cos(\theta)\\ 0 & 10-2\cos(2\theta) & 0.2\sin(\theta)\\ 0.2\cos(\theta) & 0.2\sin(\theta) & 2\cos(4\theta)\end{bmatrix}\begin{bmatrix}i_{as}\\ i_{bs}\\ i_r\end{bmatrix}$$

What is the co-energy at $i_{as} = 8$ A, $i_{bs} = 2$ A, $i_r = 6$ A and $\theta = 49°$?

W'_m[J]= 305.154 ✓ 100%

Image capture



Correct answer

W'_m [J]= 305.153662559

Submitted answer 3

yitaoma2@illinois.edu submitted at 2026-11-03 19:31:23 (CST) 100%

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Quiz4 第3题 原始页面 2/2

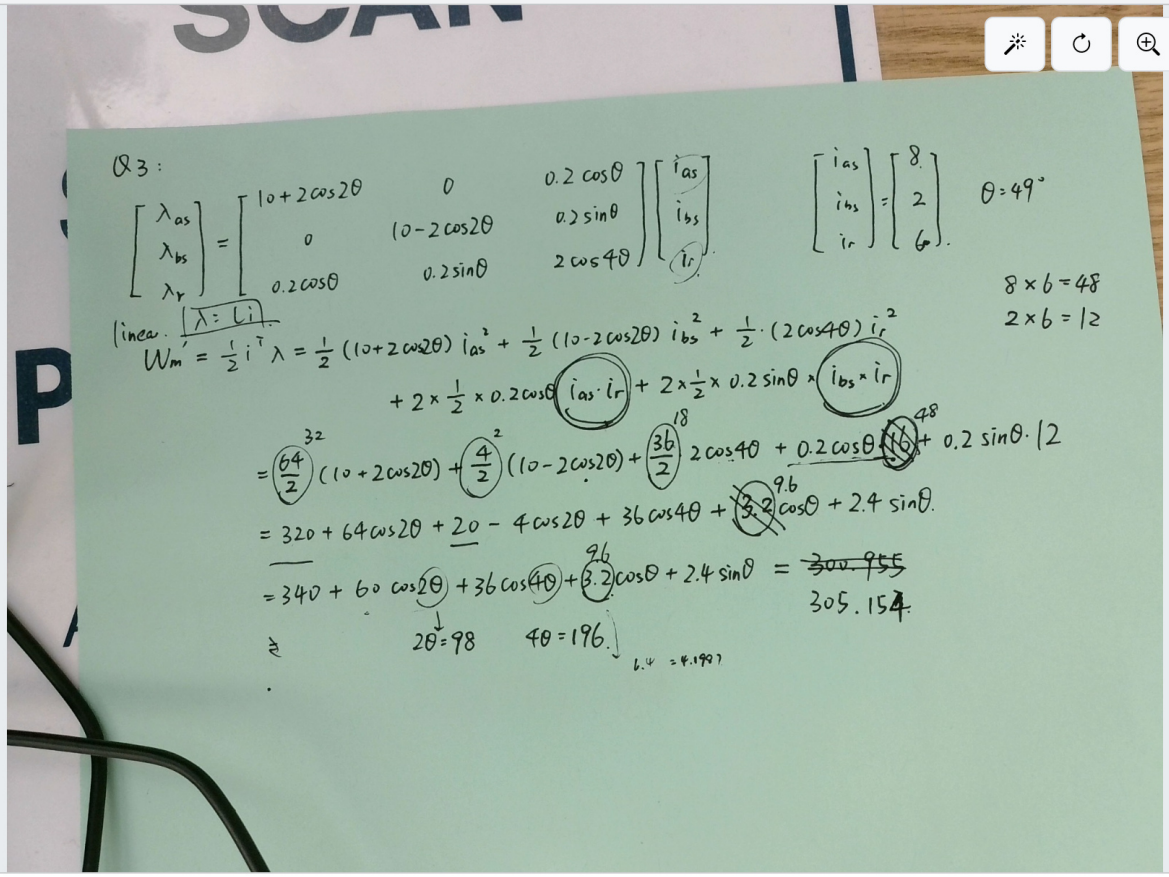
完整题目、填写值、上传回答与系统记录；相邻页有重叠

yitaoma2@illinois.edu submitted at 2026-11-03 19:31:23 (CST) 100%

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W_m' [J]= 305.154 | The correct answer used for grading was 305.1536625594601. ✓ 100%

Image capture



Submitted answer 2

yitaoma2@illinois.edu submitted at 2026-11-03 19:26:05 (CST) saved, not graded

Show

Submitted answer 1

yitaoma2@illinois.edu submitted at 2026-11-03 19:25:56 (CST) invalid, not gradable

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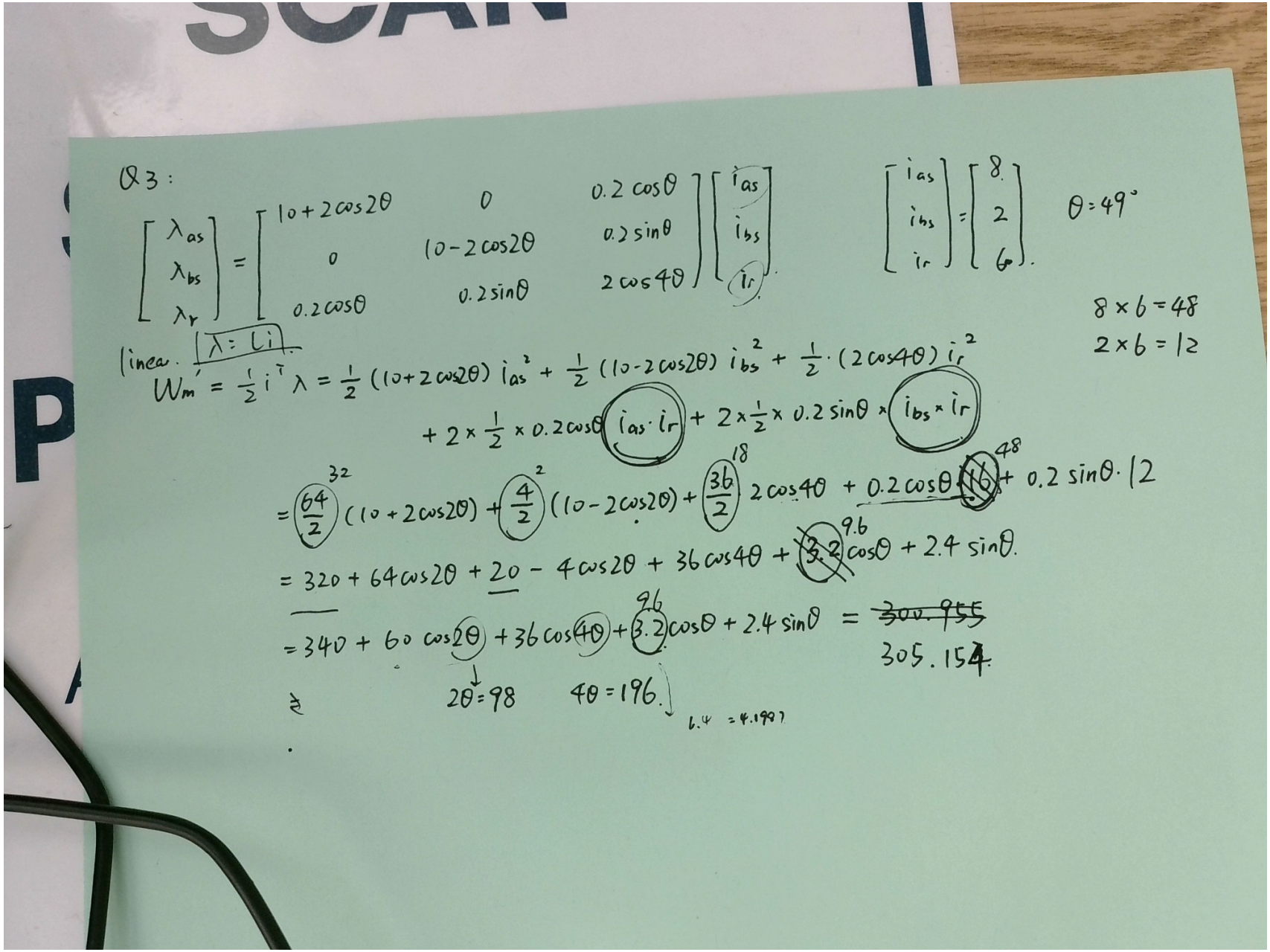
W_m' [J]= 300.955

Image capture

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Quiz4 第3题 你的完整手写回答 1/2

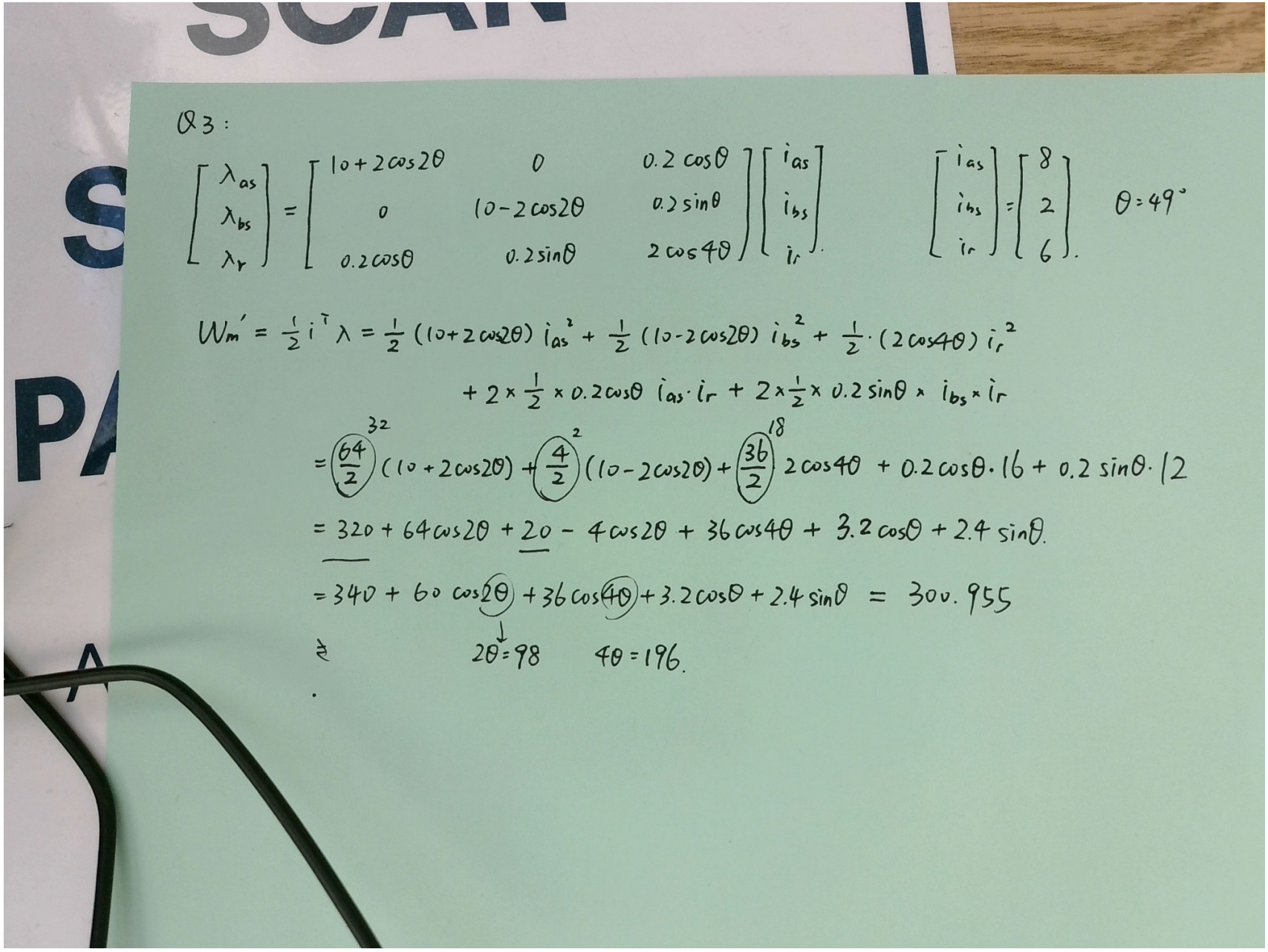
当前最终回答原图



Quiz4 Q3 | Full uploaded image | 2000 x 1500 px

Quiz4 第3题 你的完整手写回答 2/2

历史回答原图 版本2



Quiz4 Q3 | Full uploaded image | 2000 x 1500 px

第3题 三电端口的瞬时共能

中文题意

系统有三个电端口和一个机械端口，题给对称电感矩阵的对角项为 $10+2\cos 2\theta$、$10-2\cos 2\theta$、$2\cos 4\theta$；非零互感为 $0.2\cos\theta$ 和 $0.2\sin\theta$。求 $i_{as}=8\,\mathrm{A}$、$i_{bs}=2\,\mathrm{A}$、$i_r=6\,\mathrm{A}$、$\theta=49^\circ$ 时的共能 W'_m，单位 J。

从矩阵展开到数值

固定角度时，磁链对电流是线性的，且矩阵对称，因此 $W'_m=\frac{1}{2}\mathbf{i}^{\mathsf{T}}L\mathbf{i}$。对角项各带二分之一；每组非零互感在矩阵乘法里出现两次，与前面的二分之一相消。

$$
\begin{aligned}
W'_m &= \tfrac{1}{2}(10+2\cos 2\theta)i_{as}^2+\tfrac{1}{2}(10-2\cos 2\theta)i_{bs}^2 \\
&\quad +0.2i_{as}i_r\cos\theta+0.2i_{bs}i_r\sin\theta+i_r^2\cos 4\theta.
\end{aligned}
$$

把三电流和角度放回各自位置，使用 $2\theta=98^\circ$、$4\theta=196^\circ$：

$$
\begin{aligned}
W'_m &= \tfrac{1}{2}(10+2\cos 98^\circ)(8)^2+\tfrac{1}{2}(10-2\cos 98^\circ)(2)^2 \\
&\quad +0.2(8)(6)\cos 49^\circ+0.2(2)(6)\sin 49^\circ+(6)^2\cos 196^\circ \\
&= 340+60\cos 98^\circ+9.6\cos 49^\circ+2.4\sin 49^\circ+36\cos 196^\circ \\
&\approx 305.153662559\ \mathrm{J}.
\end{aligned}
$$

你的最终输入为 $305.154\,\mathrm{J}$，与系统答案一致。题目只要求此时的共能，不需要再求转矩。

最终英文作答版

At fixed θ, the flux–linkage model is linear in the currents and the inductance matrix is symmetric. Thus,

$$
\begin{aligned}
W'_m &= \tfrac{1}{2}\mathbf{i}^{\mathsf{T}}L\mathbf{i} \\
&= \tfrac{1}{2}(10+2\cos 98^\circ)(8)^2+\tfrac{1}{2}(10-2\cos 98^\circ)(2)^2 \\
&\quad +0.2(8)(6)\cos 49^\circ+0.2(2)(6)\sin 49^\circ+(6)^2\cos 196^\circ \\
&= 305.153662559\ \mathrm{J}.
\end{aligned}
$$